



Assignment 2 - Mathematics-II
Departments: CSE,ECE,ENE
Pondicherry University

👤 Aasaimani Thamizhazhagan
✉ aasaimanit@pondiuni.ac.in

📅 Apr 13, 2026 lecture
📍 M05, RSMS Building

1. Find the Laplace transform of the following:

(a) $\sin at \sinh bt$

(b) $\sin^3 2t$

(c) $e^{-2t}(2 \cos 5t - 2t \cos 2t)$

(d) $F(t) = \begin{cases} 2t & , 0 \leq t \leq 5 \\ 10 & , t > 5 \end{cases}$

(e) $e^{2t} \sin^4 t$

2. Express the following functions in terms of unit step function and find its Laplace transform:

(a) $F(t) = \begin{cases} 4 & , 0 < t < 1 \\ -2 & , 1 < t < 3 \\ 5 & , t > 3 \end{cases}$ (b) $F(t) = \begin{cases} t & , 0 < t < 2 \\ t^2 - 1 & , 2 < t < 3 \\ 5 & , t > 3 \end{cases}$

3. Find the Laplace transform of the following:

(a) $F(t) = \begin{cases} 2t & , 0 \leq t \leq 1 \\ 4 - 2t & , 1 < t \leq 2 \end{cases}$ extended periodically

4. Find the Laplace transform of

(a) i. $te^{2t} \sin 3t$

ii. $t^2 \cos at$

iii. $t(3 \sin 2t - 2 \cos 2t)$

(b) i. $\frac{\sin^2 t}{t}$

ii. $e^{-4t} \int_0^t \frac{\sin 3t}{t} dt$

(c) i. $\int_0^{t/3} \left(\frac{1 - \cos(3u)}{u} \right) du$

ii. $\int_0^t \frac{\sin^2 t}{t^2} dt$

5. Evaluate

(a) $\int_0^\infty t^2 e^{-4t} \cos t dt$ (b) $\int_0^\infty \frac{e^{-t} - e^{-3t}}{t} dt$